

NYC Traffic Collision Dashboard

Interactive Exploration of Spatial, Temporal, and Causal Crash Patterns

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Abstract

Traffic collisions are a serious concern in New York City, but raw crash tables make it difficult to see where risk concentrates, how crash patterns change over time, and which reported contributing factors appear most often. This project builds an interactive dashboard using Python, Pandas, Plotly, and Dash to support exploratory analysis of NYC motor vehicle collision data. The system combines a density map for high-risk crash locations, a ranked bar chart of contributing factors, a monthly trend chart, and summary cards controlled by shared filters for year, borough, and time of day. The design focuses on reducing overplotting, supporting comparison, and using accessible visual encodings. The final dashboard helps users move from a large dataset to concrete insights about traffic safety patterns in NYC.

1 Introduction and Motivation

New York City has one of the most complex urban transportation environments in the United States. Cars, buses, taxis, bicycles, pedestrians, delivery vehicles, and emergency vehicles all share dense street networks. Because of this complexity, traffic collision data is not only large but also difficult to interpret directly. A spreadsheet of crash records can show what happened in individual incidents, but it does not easily answer higher-level questions such as:

- Where are serious crash hotspots located?
- Which reported contributing factors dominate the dataset?
- How do crash counts change by month, borough, or time of day?
- Can users compare subsets without manually writing code?

Problem definition. The main problem addressed in this project is the transformation of a large, event-level traffic collision dataset into an interactive visual system that supports spatial, temporal, and causal exploration.

Why it matters. Traffic safety is a public concern. Visualizing collision patterns can help analysts, students, and community members identify high-risk areas, understand recurring causes, and discuss where interventions may be needed. Even when the dashboard is exploratory rather than predictive, it can make patterns visible that are hard to see from raw tables.

Target users. The primary audience includes urban safety analysts, transportation researchers, students, and the general public. The intended use case is exploratory analysis: a user selects a year, borough, and time window, then examines how the map, causes, monthly trend, and summary statistics change together.

2 Related Work and Background

The project is informed by several visualization ideas from the information visualization literature: task abstraction, perceptual effectiveness, visual encoding, interaction, and the challenge of representing large point datasets.

2.1 Visualization design principles

Munzner argues that visualization design should begin with understanding the task, data abstraction, and intended user goals rather than choosing chart types first [2]. This project follows that idea by starting from four analytical tasks: locating hotspots, comparing time periods, ranking causes, and filtering by subset. The dashboard views are then chosen to support those tasks directly.

Ware emphasizes the role of human perception in visualization design, especially the importance of visual channels such as position, color, and size [3]. In this dashboard, position is used for the most precise comparisons in the bar and line charts, while color is used mainly to support grouping, emphasis, and density. This avoids relying on color alone for critical interpretation.

Tufte’s work on graphical integrity and data-ink ratio is also relevant [4]. The dashboard avoids unnecessary decorative elements and keeps the main focus on the data views. Summary cards, chart titles, and filters are included because they help interpretation, not because they add visual complexity.

2.2 Interactive exploration

Shneiderman’s visual information-seeking mantra—“overview first, zoom and filter, then details-on-demand”—is a useful way to evaluate this dashboard [5]. The dashboard provides overview-level summaries through cards and charts, filtering through dropdown controls, and details-on-demand through Plotly hover interactions.

Heer, Bostock, and Ogievetsky describe interactive visualization for the web as a way to support exploratory thinking through rapid view updates and direct manipulation [6]. This project applies that approach in a small but focused Dash application: the user changes filter controls and all views update together.

2.3 Maps, overplotting, and density

Collision records are geographic point data. A raw point map is tempting, but in NYC it quickly becomes unreadable because many crashes occur in dense street networks. This is an overplotting problem: many marks overlap, hiding local patterns and making the map look saturated.

Kernel density estimation is a common approach for showing spatial concentration instead of every individual point [7]. The dashboard uses a density map for serious crashes so the user can see clusters rather than thousands of overlapping dots. This is a deliberate tradeoff: the map no longer shows every crash as a separate mark, but it better supports the hotspot-detection task.

2.4 Color accessibility

Color is a major part of visualization design, but it can also create accessibility problems. Brewer’s work on color schemes is widely used in cartography and visualization because it emphasizes sequential, diverging, and qualitative palettes for different data types [8]. This project uses a perceptually ordered sequential colorscale for map density and a colorblind-conscious qualitative palette for the cause chart. Labels and ordering remain present so the dashboard is not dependent on color alone.

3 Data Description

The dataset used is NYC Open Data’s *Motor Vehicle Collisions – Crashes* dataset [1]. The raw dataset contains crash-level records with date, time, location, borough, injury/fatality counts, vehicle information, and reported contributing factors.

3.1 Fields used

The dashboard uses the following fields:

Table 1: Main data fields used in the dashboard.

Field	Purpose in dashboard
Crash date and crash time	Year, month, and time-of-day filtering
Borough	Geographic filtering and comparison
Latitude and longitude	Density map locations
Persons injured / killed	Summary cards and serious-crash map filtering
Contributing factor vehicle 1	Ranked crash-cause bar chart
Vehicle type code 1	Kept in cleaned data for possible extension

3.2 Preprocessing

The preprocessing script `src/clean_data.py` performs the following steps:

1. Reads selected columns from the raw collision CSV.
2. Parses `CRASH DATE` and extracts year, month, month name, and hour.
3. Removes rows with missing dates or missing coordinates.
4. Keeps only NYC-like coordinate bounds (latitude 40.45–40.95 and longitude -74.30– -73.65).
5. Keeps records from 2022 onward to improve performance and keep the analysis recent.
6. Fills missing borough values as `UNKNOWN`.
7. Fills missing contributing factors as `Unspecified`.
8. Writes a cleaned CSV to `data/processed/collisions_cleaned.csv`.

After cleaning, the project dataset contains **369,660 rows** from **2022 through April 10, 2026**. The default dashboard view currently starts at the most recent year, 2026. For 2026, the cleaned data contains **21,177 crashes**, **11,235 injured persons**, and **49 fatalities**.

3.3 Data challenges

The dataset presents several visualization and analysis challenges:

- **Scale:** hundreds of thousands of cleaned rows can slow interactive rendering if all points are drawn directly.
- **Missingness:** some records have unknown boroughs or unspecified contributing factors.
- **Overplotting:** raw crash points overlap heavily in dense areas of NYC.
- **Interpretation limits:** reported contributing factors should not be treated as perfectly causal; they are recorded labels in the crash report.
- **Exposure not included:** crash counts are not normalized by traffic volume, population, or road length.

4 Design and Methodology

4.1 Analytical tasks

The dashboard is organized around four main analytical tasks:

1. **Spatial task:** locate high-risk areas where serious crashes cluster.
2. **Causal/categorical task:** identify the most common reported contributing factors.
3. **Temporal task:** observe monthly crash trends.
4. **Comparison task:** compare different years, boroughs, and times of day through shared filters.

4.2 Dashboard layout

The layout uses a top-down structure:

1. Title and short project description.
2. Filter controls for year, borough, and time of day.
3. Summary cards for total crashes, injuries, fatalities, and most common cause.
4. Main visualization grid: map on the left; cause and trend charts on the right.

This structure creates a clear hierarchy. The user first understands the dashboard topic, then selects a subset, then reads summary statistics, and finally explores detailed views.

4.3 Visualization choices

Density map. The map answers: “Where are high-risk collision areas?” A density map was selected instead of a raw scatter plot because raw points produce severe overplotting in NYC. The current map filters to crashes with **3+ injuries or any fatality**. This makes the map focus on serious incidents instead of showing that collisions happen almost everywhere in the city.

Horizontal bar chart. The cause chart uses horizontal bars because contributing-factor labels are long. Position along a common x-axis is an effective channel for comparing magnitudes. The bars are sorted and labeled with counts, so users can read values even if colors are hard to distinguish.

Line chart. The monthly trend chart uses a line with markers and subtle area fill. This supports temporal comparison and lets users see whether crash counts rise, fall, or remain stable across months.

Summary cards. Summary cards provide immediate quantitative context before users inspect the charts. They also help prevent misinterpretation: when filters are changed, the user can see the size and severity of the selected subset.

4.4 Color and accessibility

The abstract feedback specifically warned to “watch out for color encoding” because similar colors can reduce clarity. The final design responds to that feedback in several ways:

- The density map uses **Cividis**, a perceptually ordered sequential colorscale that remains more readable for colorblind and grayscale viewing than many rainbow-style palettes.
- The cause chart uses a colorblind-conscious qualitative palette inspired by Okabe–Ito colors.
- Bar values are written as text labels, so the chart does not rely on color alone.
- Summary cards use both color and text labels; color is supportive rather than the only encoding.
- The dashboard uses strong typographic hierarchy and spacing so visual grouping is not dependent only on color.

4.5 Interaction design

The dashboard provides three primary filters:

- **Year:** selects one year from 2022–2026.
- **Borough:** filters to a borough or all boroughs.
- **Time of day:** filters to morning, afternoon, evening, night, or all day.

These filters update every view through a Dash callback. This makes the system a coordinated multiple-view dashboard rather than a collection of disconnected charts.

4.6 Alternative designs considered

Several alternatives were tested or considered:

- **Raw point map:** rejected because dense crash points cluttered the map and hid patterns.
- **Severity-weighted density:** initially considered, but it made the colorbar confusing because many individual records had low severity while the density surface became saturated.
- **All-injury crash density:** rejected because it made the city look uniformly high-risk. Filtering to 3+ injuries or fatalities gives a more meaningful hotspot view.
- **Stacked/clustered cause chart by borough:** rejected for the main dashboard because it would make the cause chart harder to read. Borough comparison is handled through filters.

5 System Description and Implementation

Tools used. The system is implemented in Python using Pandas for data processing, Plotly Express for visualizations, and Dash for the web dashboard. The project dependencies are listed in `requirements.txt`.

5.1 Architecture

The project has two main code components:

- `src/clean_data.py`: reads the raw CSV, filters/cleans fields, creates derived time variables, and writes the processed dataset.
- `app.py`: loads the processed dataset, defines the dashboard layout, filters data based on user input, creates figures, and updates the app through a Dash callback.

5.2 Callback workflow

The app follows this workflow:

1. User changes year, borough, or time-of-day dropdown.
2. `filter_data()` applies the selected subset.
3. `create_summary_cards()` recomputes key totals.
4. `create_map()`, `create_cause_chart()`, and `create_trend_chart()` regenerate visualizations.
5. The Dash callback returns all updated components together.

5.3 Performance choices

The system keeps performance manageable by using a recent-year cleaned dataset and by sampling the map input when necessary. The map view is also restricted to serious crashes, which improves both responsiveness and interpretability.

6 Results and Insights

The dashboard reveals several concrete patterns in the cleaned data. The following findings use the default 2026 view (all boroughs, all day), which contains 21,177 crashes.

6.1 Insight 1: Driver inattention/distraction dominates reported causes

The top reported contributing factor in 2026 is **Driver Inattention/Distracted**, with **4,884 crashes**. This is much higher than the second-ranked factor, **Failure to Yield Right-of-Way**, with **1,265 crashes**. The skew suggests that a small number of reported factors account for a large share of crashes.

Table 2: Top reported contributing factors in 2026.

Reported contributing factor	Crashes
Driver Inattention/Distracted	4,884
Failure to Yield Right-of-Way	1,265
Following Too Closely	1,016
Other Vehicular	874
Passing or Lane Usage Improper	750
Passing Too Closely	744
Backing Unsafely	726
Unsafe Speed	614

6.2 Insight 2: Serious-crash hotspots are a smaller subset

For 2026, the serious-crash map filter (3+ injuries or any fatality) selects **680 crashes**, compared with 21,177 total crashes. This is about **3.2%** of all 2026 crashes. This validates the design decision to focus the map on serious crashes: plotting every injury crash made the city appear uniformly saturated, while the serious-crash filter supports meaningful hotspot analysis.

6.3 Insight 3: Brooklyn has the highest known borough crash count

In the 2026 cleaned data, the largest borough count is **Brooklyn** with **5,748 crashes**. Queens has **4,158**, Manhattan has **2,919**, the Bronx has **2,665**, and Staten Island has **696**. There are also **4,991** records with UNKNOWN borough, which is a data-quality limitation and should be considered when interpreting borough comparisons.

6.4 Insight 4: Afternoon has the highest crash count by time-of-day

Using the dashboard’s time-of-day bins for 2026, afternoon (12 PM–4 PM) has the highest crash count with **5,973 crashes**. Morning has **5,499**, night has **5,254**, and evening has **4,451**. This suggests that crash activity is not limited to late-night or commute periods; daytime traffic conditions also matter.

6.5 Insight 5: 2026 data is partial

The 2026 data runs through April 10, 2026. Monthly counts are January **6,548**, February **6,268**, March **6,511**, and April **1,850**. The April drop should not be interpreted as a true decline because April is incomplete in the dataset.

6.6 Screenshots to include

The following screenshots should be added before final PDF submission. The LaTeX is already prepared to include them if they are saved with these filenames:

- `figures/dashboard_overview.png`: full dashboard with filters, summary cards, and all three views.
- `figures/map_hotspots.png`: close-up of “Where are high-risk collision areas?”
- `figures/cause_factors.png`: close-up of “What causes most reported crashes?”
- `figures/monthly_trend.png`: close-up of “How do crashes change by month?”
- `figures/filter_example.png`: example filtered view (for example, Brooklyn + Afternoon) to demonstrate interaction.

Missing figure: `figures/dashboard_overview.png`

Add a full dashboard screenshot showing the title, filters, summary cards, map, cause chart, and trend chart.

Figure 1: Full dashboard overview.

Missing figure: `figures/map_hotspots.png`

Add a screenshot of the “Where are high-risk collision areas?” density map.

Figure 2: High-risk density view for serious crashes.

Missing figure: `figures/cause_factors.png`

Add a screenshot of the ranked contributing-factor bar chart.

Figure 3: Top reported contributing factors.

Missing figure: `figures/monthly_trend.png`

Add a screenshot of the monthly crash trend chart.

Figure 4: Monthly crash trend.

Missing figure: `figures/filter_example.png`

Add a screenshot of one filtered state, such as Brooklyn + Afternoon, to demonstrate interaction.

Figure 5: Example filtered dashboard state demonstrating interaction.

7 Evaluation

A full formal user study was outside the project scope, but the dashboard was evaluated through a lightweight design review based on the rubric requirements.

- **Functionality:** filters update all views; charts render from the cleaned dataset.
- **Perceptual clarity:** position is used for precise comparisons in the bar and line charts; density is used for spatial concentration.
- **Color accessibility:** the map uses a perceptually ordered colorscale; bar labels provide redundant encoding.
- **Interaction:** year, borough, and time-of-day filters support meaningful analytical comparisons.
- **Reproducibility:** dependencies are listed in `requirements.txt`, and the app can be run with `python app.py`.

8 Limitations and Future Work

Limitations.

- Counts are not normalized by traffic exposure, population, or roadway length.
- Some records have unknown boroughs or unspecified contributing factors.
- Contributing factors are reported labels and should not be interpreted as definitive causal proof.
- The 2026 data is partial, so monthly trends for 2026 must be interpreted carefully.
- The dashboard focuses on exploration, not prediction or statistical modeling.

Future work.

- Add traffic-volume or road-network normalization to estimate risk rates rather than raw counts.
- Add side-by-side borough comparison instead of requiring users to switch filters manually.
- Add a small user study to evaluate whether users can answer key questions quickly.
- Add more contextual layers, such as bike lanes, school zones, or Vision Zero priority areas.
- Extend the dashboard to compare multiple years at once.

9 Conclusion

This project created an interactive dashboard for exploring NYC motor vehicle collisions through spatial, temporal, and categorical views. The system uses a density map for serious crash hotspots, a ranked bar chart for contributing factors, a monthly trend chart, and summary cards connected through shared filters. The design responds to key visualization challenges in the dataset: scale, overplotting, missing values, and color accessibility. The dashboard demonstrates how information visualization can transform a large civic dataset into a clearer, more usable tool for understanding traffic safety patterns.

References

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- [2] Munzner, T. *Visualization Analysis and Design*. CRC Press, 2014.
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